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# NOTES ON A DEGENERATE PARABOLIC EQUATION

## Based on *Uniqueness and Root-Lipschitz Regularity for a Degenerate Heat Equation*

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# 1 Introduction

[Heat Equation and Adjoint equation] The heat equation

$$\partial_t u = \Delta u, \quad u \in \mathbb{R}^n \times \mathbb{R}$$

has a class of solution

$$\Gamma(x, t; \xi, \tau) = \frac{1}{(2\sqrt{\pi})^n}(t-\tau)^{-n/2} \exp\left(-\frac{|x-\xi|^2}{4(t-\tau)}\right), \quad (\xi, \tau) \in \mathbb{R}^n \times \mathbb{R}$$

where for any continuous function $f(x) \lesssim \exp\left(h|x|^2\right)$ for some $h > 0$, then the integral

$$u(x, t) = \int \Gamma(x, t; \xi, 0) f(\xi) d\xi$$

exists for $0 < t \leq T$ if $T < 1/4h$ and $u(x, t) \to f(x), t \to 0$, and $\Gamma$ is called a fundamental solution of the heat equation; and $u$ is a solution of the Cauchy problem, i.e. the problem of finding a solution to the heat equation for $0 < t \leq T$ with the initial condition $u(x, 0) = f(x)$.

The adjoint equation if defined as

$$\partial_t v = -\Delta v, \quad v \in \mathbb{R}^n \times \mathbb{R}$$